Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/10/2006
PI Name:	Hangarter, Roger P Ph.D.		
Project Title:	Gravity response mechanisms of lateral organs and the control of plant architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	47405-7107	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mullen, Jack (Indiana University)		
Grant/Contract No.:	NNA04CC55G		
Performance Goal No.:			
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Task Description:	We previously demonstrated that the orientation of lateral organs in plants (roots, branches and leaves) is determined by what is called the gravitropic set-point angle (GSA). That is, the specific orientation of lateral organs is genetically determined but controlled by the gravitropic responses of the lateral organs. The goal of this project is to use a combination of genetic, molecular, and physiological approaches to better understand the role of gravity in guiding whole plant development. Specifically, we are using Arabidopsis as a model system for identifying genes required for controlling the GSA of later roots, rosette leaves and inflorescence branches. We have isolated a number of mutants that display altered GSA responses in lateral organs. The mutants have been designated as mgsa mutants for modified gravitropic set-point angles. The molecular and physiological analyses of the mgsa mutants will provide new insight into the basis of GSA regulation and, thus, a better understanding of the physiological and molecular mechanisms by which plants use gravity to control their architecture. To discover and characterize genes that are required for controlling the orientaion of lateral organs during plant development and to understand the mechanisms by which plants sense and respond to gravity. The molecular and physiological analyses of the gsa mutants will provide new insight into the basis of GSA regulation and, thus, a better understanding of the physiological and molecular mechanisms by which plants use gravity to control their architecture. Since the bulk of a plant consists of lateral organs, our studies will contribute to a better understanding of the fundamental developmental biology of plant form and may eventually lead to the development of improved plant varieties for growth on future long-term space missions or for agriculture.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since the bulk of a plant consists of lateral organs, our studies will contribute to our understanding of the fundamental developmental biology of plant form and may eventually lead to the development of improved plant varieties for high-density agricultural.
Task Progress:	We have continued to identify Arabidopsis mutants that have altered lateral root orientation while maintaining normal primary root orientation. Likewise, we have identified mutants that have altered leaf and/or branch orientation. All of these mutants retain normal gravitropic responses in the primary roots and primary shoot. Moreover, the lateral organs also display a robust gravitropic response. The mutations identified appear to specifically affect the GSA of the lateral organs, causing them to orient differently than in wild-type Arabidopsis plants. We have designated the mutants as mgsa mutants for modified gravitropic set-point angles. We have made significant progress towards the molecular identification of the MGSA4 gene. Plants mutant for mgsa4, orient their leaves and inflorescence branches in a much more vertical position than wild-type plants. As a result the mutant has a significantly different architecture in the above-ground parts of the plant. Through fine-scale genetic mapping of mgsa4, we have narrowed the mutant locus to a region on chromosome 4. Transformation of mgsa4 mutant plants with wild-type DNA from BACS that are located in the region of mgsa4 have yielded plants that have wild-type leaf and branch orientations indicating the plants have been transformed with a wild-type copy of the MGSA4 gene. Although we have not yet identified the specific gene responsible for rescuing the mutant, we have been able to narrow the list of candidates to about 10 genes. We have also mapped the mgsa3 mutation to a region on chromosome 1. The lateral roots of the mgsa3 grow more horizontally than in wild-type plants. As part of our analysis of the GSA response of lateral organs, we have carried out a detailed analysis of the GSA response in wild-type Arabidopsis plants and several other plant species using time-lapse methods. Our results demonstrate that the orientation of lateral organs in diverse species is primarily determined by the gravity responses of the lateral organs. Our results also demonstrate that the GSA response changes with the developmental stage of the lateral organs. Based on these physiological studies, we conclude that the overall architecture of a plant results from the orchestrated developmental changes in the GSA of the lateral organs. Moreover, the GSA can be modified by environmental stimuli such as the quality (red/far-red ratio) and quantity of light. A paper describing the effect of light on leaf orientation is in press and a more detailed manuscript describing the relationship between lateral organ orientation and plant architecture is in preparation.
Bibliography Type:	Description: (Last Updated: 08/11/2025)
Articles in Peer-reviewed Journals	Mullen JL, Wolverton C, Hangarter RP. "Apical control, gravitropic signaling, and the growth of lateral roots in Arabidopsis." Adv Space Res. 2005;36(7):1211-7. , Jul-2005
Articles in Peer-reviewed Journals	Mullen JL, Weinig C, Hangarter RP. "Shade avoidance and the regulation of leaf inclination in Arabidopsis." Plant, Cell & Environment, In Press, January 2006. , Jan-2006
Presentation	Mullen, J.L.; Hangarter, R.P. "Gravity responses in lateral organs of Arabidopsis" Annual meeting, American Society for Gravitational and Space Biology, Reno, NV Nov-2005
Presentation	Mullen, J.L.; Wolverton, C.; Hangarter, R.P. "Role of auxin in the control of lateral root growth in Arabidopsis" Annual meeting, American Society of Plant Biologists, Seattle, WA Jul-2005
Presentation	Mullen, J.L.; Hangarter, R.P. "Shape shifting in plants: gravitropism and plant form" Plant Biology 2004, Annual Meeting of the American Society of Plant Biologists. Orlando, FL Jul-2004
Presentation	Hangarter, R.P. "Gravity responses of lateral organs in shaping plant architecture" Dept of Plant Biology, UC Berkeley, Oct 3 Oct-2005
Presentation	Hangarter, R.P. "Light and gravity as key regulators of plant architecture" Pennsylvania State University, Oct 12 Oct-2004
Presentation	Hangarter, R.P. "Light, gravity and the regulation of plant architecture" Pioneer Hi-Bred International, Inc. Sep-2004

Presentation	Mullen, J.L.; Hangarter, R.P. "Gravity response mechanisms of lateral organs and the control of plant architecture in Arabidopsis." Annual meeting, American Society for Gravitational and Space Biology, New York Nov-2004
Presentation	Mullen, J.L.; Wolverton, C.; Hangarter, R.P. "Apical control, gravitropic signaling and the growth of lateral roots in Arabidopsis" COSPAR, Paris, France Jul-2004